

Data Flow Diagram

Date	5 July 2026
Team ID	xxxxxx
Project Name	AI Learning Support Assistant
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

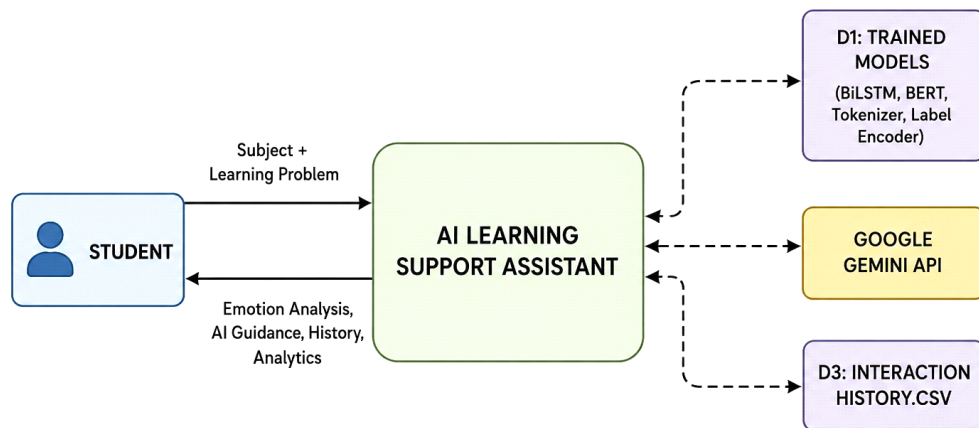
DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

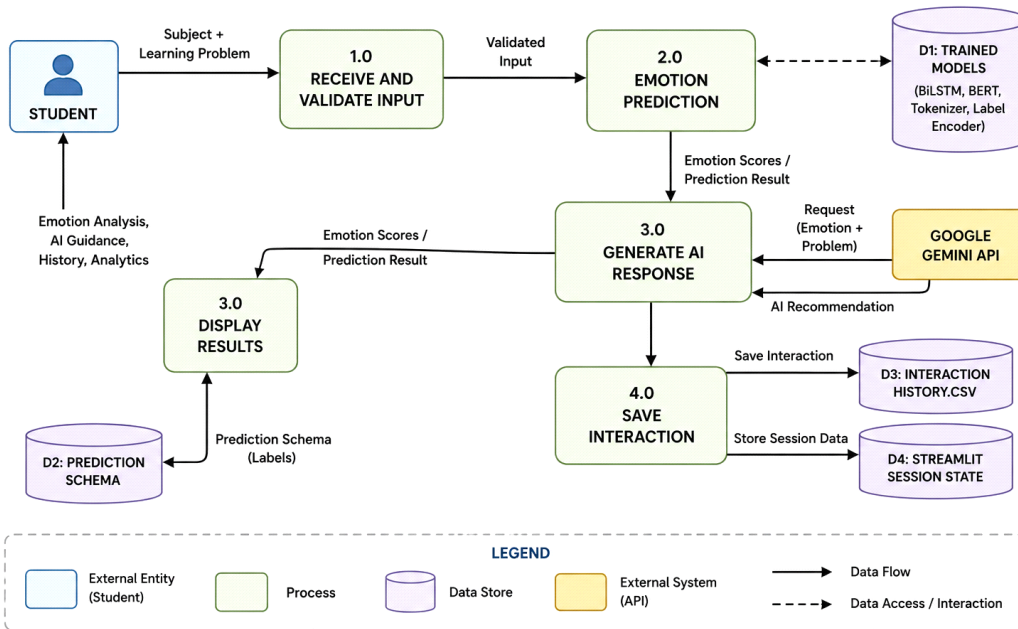
Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

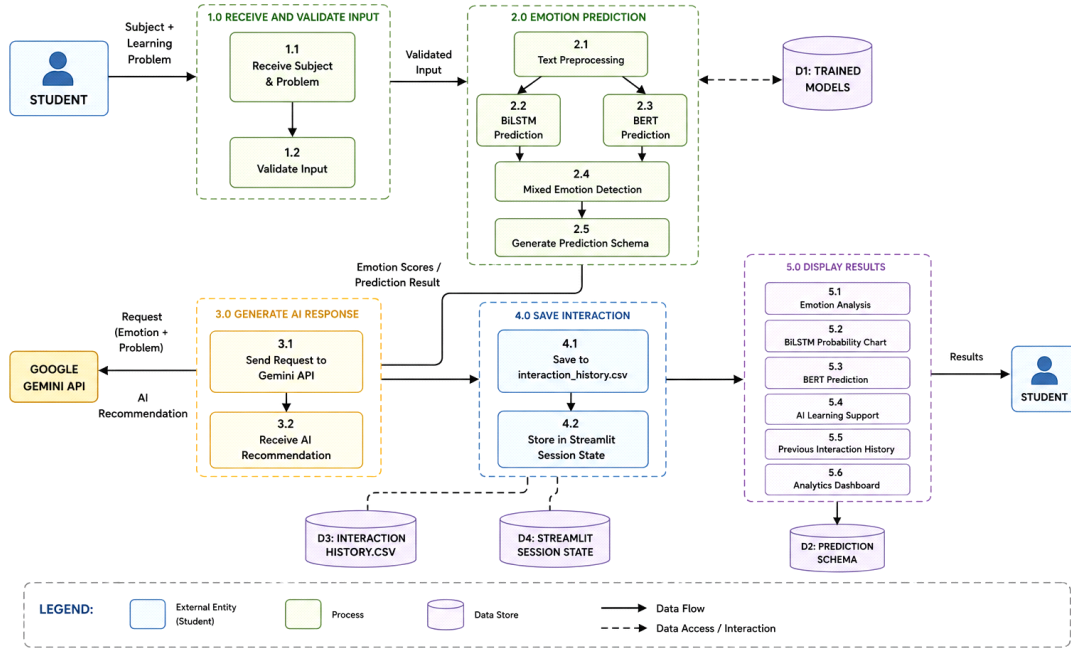
1. CONTEXT DIAGRAM (LEVEL-0 CONTEXT)



2. LEVEL-1 DATA FLOW DIAGRAM (DFD) AI LEARNING SUPPORT ASSISTANT

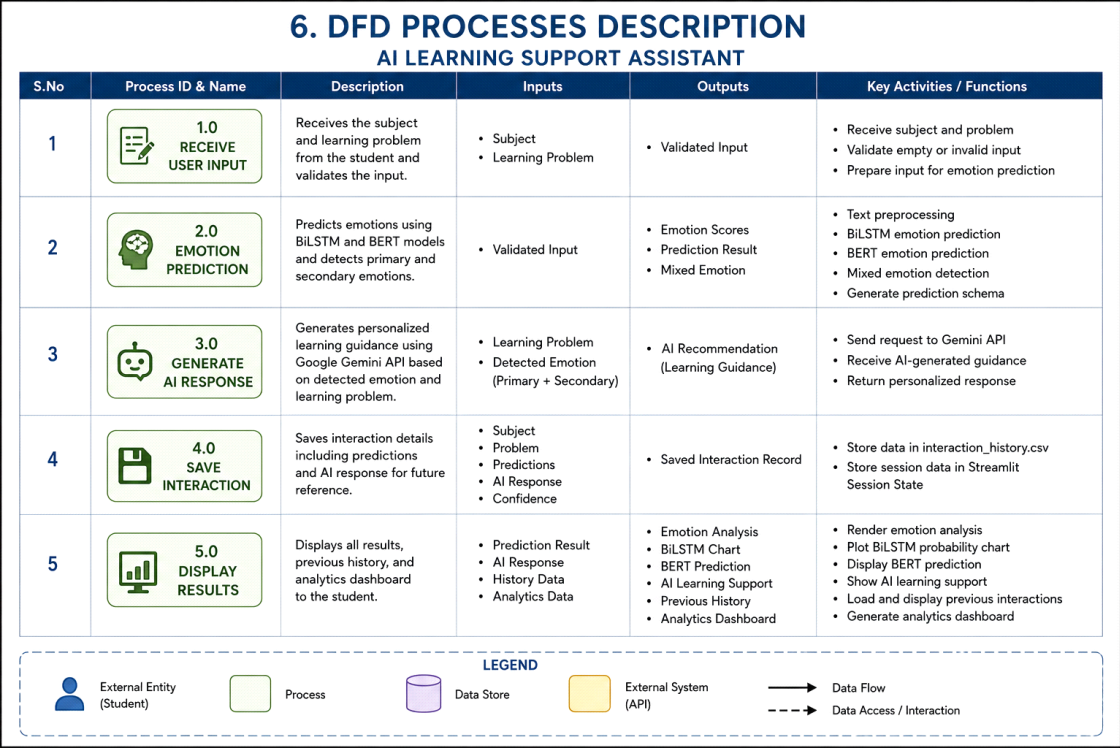
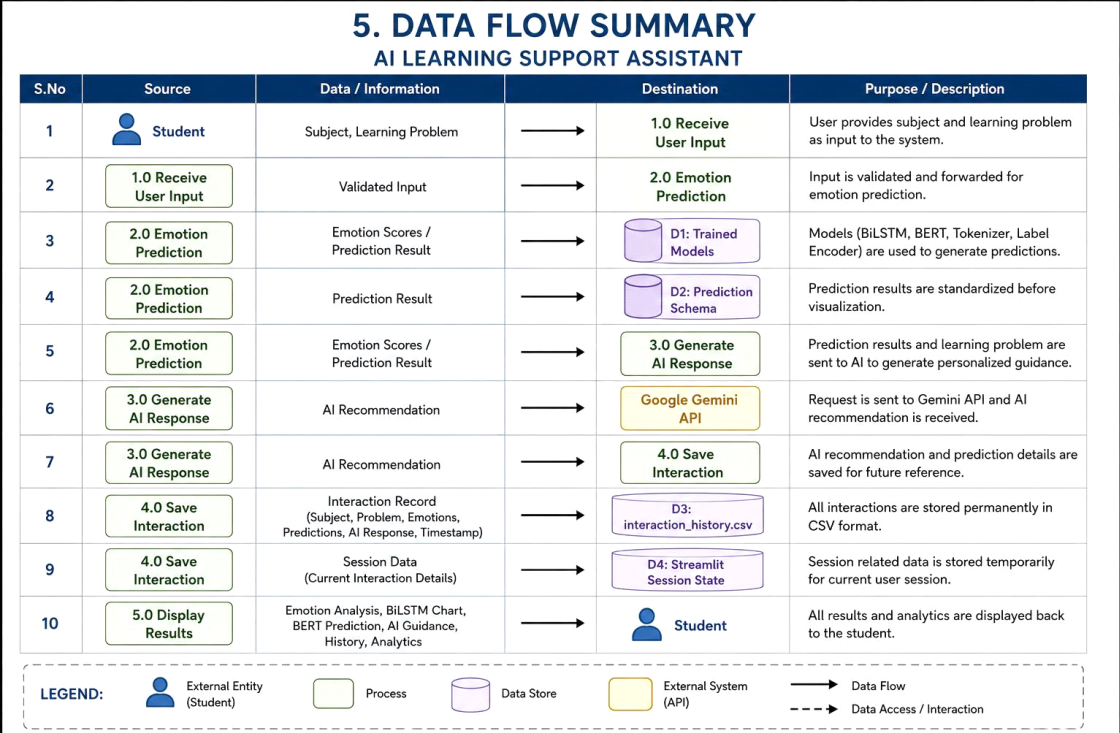


3. LEVEL-2 DATA FLOW DIAGRAM (DFD) AI LEARNING SUPPORT ASSISTANT



4. DFD COMPONENTS AI LEARNING SUPPORT ASSISTANT

Component Type	Component Name / ID	Description	Input	Output	Stored In / Connected To
EXTERNAL ENTITY  Student	Student	The user of the system who submits learning problems and views results, history, and analytics.	- Subject - Learning Problem	- Emotion Analysis - AI Learning Guidance - Previous History - Analytics Dashboard	Interacts with Process 1.0 (Receive User Input)
	1.0 Receive User Input	Receives subject and learning problem from the student and validates the input.	- Subject - Learning Problem	Validated Input	Sends validated input to Process 2.0 (Emotion Prediction)
PROCESSES 	2.0 Emotion Prediction	Predicts emotions using BiLSTM and BERT models and detects mixed emotions.	Validated Input	- Emotion Scores - Prediction Result - Mixed Emotion	Uses D1: Trained Models and updates D2: Prediction Schema
	3.0 Generate AI Response	Generates personalized learning guidance using Google Gemini API.	- Learning Problem - Detected Emotion (Primary + Secondary)	AI Recommendation (Learning Guidance)	Connects with Google Gemini API
	4.0 Save Interaction	Saves the interaction details including predictions and AI response.	- Subject - Problem - Predictions - AI Response - Confidence	Saved Interaction Record	- Stores in D3: interaction_history.csv - Stores in D4: Streamlit Session State
	5.0 Display Results	Displays all results, history, and analytics to the student.	- Prediction Result - AI Response - History Data - Analytics Data	- Emotion Analysis - BiLSTM Chart - BERT Prediction - AI Learning Support - Previous History - Analytics Dashboard	Reads from D2, D3, D4 and displays to Student
	D1: Trained Models	Stores trained BiLSTM model, BERT model, tokenizer, and label encoder.	–	Provides models for emotion prediction	Connected to Process 2.0 (Emotion Prediction)
DATA STORES 	D2: Prediction Schema	Stores standardized prediction results before visualization.	Prediction Result	Standardized Emotion Result	Used by Process 5.0 (Display Results)
	D3: interaction_history.csv	Stores all user interactions permanently in CSV format.	Interaction Record	Saved Interaction History	Used by Process 4.0 and read by Process 5.0
	D4: Streamlit Session State	Stores current session data, recent interactions, and dashboard information.	Session Data	Session Related Data	Used by Process 4.0 and read by Process 5.0
	Google Gemini API	Generates personalized learning guidance based on detected emotion and learning problem.	Request (Learning Problem + Detected Emotion)	AI Learning Recommendation	Connected to Process 3.0 (Generate AI Response)



Data Flow Summary

Source	Data	Destination
Student	Subject, Learning Problem	Receive User Input
Receive User Input	Validated Input	Emotion Prediction
Emotion Prediction	Emotion Scores	Trained Models / Prediction Schema
Prediction Schema	Emotion Result	Gemini AI
Gemini AI	AI Learning Guidance	Save Interaction
Save Interaction	Interaction Record	CSV History
Save Interaction	Session Data	Streamlit Session State
CSV + Session State	History Data	Display Results